

Real-time object detection and walking guidance using XR glasses and AI

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Abstract—This study proposes a real-time assistive system that integrates XR glasses and AI-based object detection technology to support the mobility and safety of elderly and disabled individuals. The system detects environmental objects and obstacles using a YOLO-based AI model and delivers visual and auditory alerts through an XR interface. It provides indoor and outdoor navigation assistance, real-time obstacle warnings, and remote caregiver support. Field evaluations indicate the system enhances independent walking and situational awareness, suggesting its potential as a core technology for inclusive smart mobility environments.

Keywords—xr glasses, object detection, assistive technology, pedestrian safety, yolov8, smart mobility

I. INTRODUCTION

Our society has already entered an aging society, and assistive technology is becoming increasingly important to improve the quality of life and social participation of older adults and people with disabilities. The elderly and people with disabilities face many limitations in mobility and communication, and various technological supports to overcome them are essential. In particular, people with visual, hearing, and intellectual disabilities and the elderly face difficulties in mobility and communication in daily life and social activities, and existing assistive technologies have limitations in providing customized services and processing real-time information.

Recognition of these issues is spurring a society-wide response, particularly at the government and local government level, with increased policy efforts to ensure the right to walk and access to information for people with disabilities. Improving the mobility and digital accessibility of older adults and people with disabilities is more than just a matter of convenience; it is now seen as a key factor in social inclusion and independent living, and the need for technological responses to keep pace.

To address these issues, this research proposes a system that fuses the latest extended reality (XR) and artificial intelligence (AI) technologies to fundamentally improve mobility and communication issues for the elderly and disabled. XR glasses can provide digital information in real-time to the

user's real-world environment, including augmented reality (AR) and mixed reality (MR), and AI can provide adaptive information in real-time by performing environment recognition, object recognition, hazard detection, etc.

The purpose of this research is to utilize an artificial intelligence system based on XR glasses to enable the elderly and disabled to live more safely and independently. To this end, we aim to remove barriers to mobility and communication through the integration of indoor and outdoor transportation guides, digital communication support, and contactless assistant services, which are the core components of the system. The system is expected to contribute to the realization of an inclusive society by not only solving problems of daily mobility and social interaction, but also facilitating the social and economic participation of people with disabilities and the elderly.

II. RELATED TECHNOLOGIES AND RESEARCH TRENDS

Recent technological advances have led to the rapid development of extended reality (XR) and artificial intelligence (AI) technologies, and attempts are being made to apply these technologies to various fields. In particular, XR technologies, which include augmented reality (AR), virtual reality (VR), and mixed reality (MR), are revolutionizing user experience by enabling users to interact between physical and digital environments.

Existing assistive devices to support older adults and people with disabilities are primarily based on limited functionality such as voice guidance, haptic feedback, and obstacle detection. Examples include smart canes, tactile interface devices, and voice-based navigation systems, but many of these devices rely on a single sensory stimulus or provide a limited range of information, limiting their ability to comprehensively recognize dynamic changes in the environment or complex hazards. In addition, the lack of real-time adaptive guidance capabilities due to the system structure based on static data or preset paths has been pointed out as a major limitation. Against this backdrop, the need for assistive systems that can recognize complex environments and respond to situations is becoming increasingly important.

Advances in artificial intelligence technology have opened up the possibility of overcoming these limitations by enabling real-time recognition and processing of the surrounding environment. Recent advances in deep learning-based object recognition and contextual awareness technologies have led to a growing number of applications in the field of smart assistive devices for the elderly and disabled. For example, AI-based object detection technology has been effectively used to help users move safely in obstacle-rich environments and provide customized information.

Internationally, there are many examples of AI and XR technology convergence assistive devices, such as Cognixion's AI-based accessibility tools to help people with disabilities communicate independently and control smart homes, and Ekso Bionics' EksoNR to support neurorehabilitation therapy through wearable robotics.

Based on these previous studies, this research aims to propose a comprehensive and real-time support system for the elderly and disabled that fuses XR and AI technologies. By doing so, it is expected to develop an innovative system that goes beyond the existing limitations technologically and actively promotes social participation and independent living for the disabled and elderly.

III. OVERVIEW OF THE PROPOSED XR SYSTEM

The XR system proposed in this study aims to support the mobility and communication of the elderly and disabled by fusing the latest XR technology with artificial intelligence (AI). The proposed system has an interlocking structure consisting of XR glasses, a mobile terminal, and a server, which enables users to effectively interact with their surroundings in real time.

As a wearable device, XR glasses play a key role in effectively bringing digital information into the user's real world. The XR glasses worn by the user acquire real-time images of the surrounding environment through built-in cameras and sensors, and the acquired images are transmitted to a mobile terminal directly connected via a wired cable.

The mobile terminal performs AI-based object recognition and environmental analysis based on the transmitted video data. The results are fed back to the XR glasses, where real-time feedback is provided to the user in the form of visual overlays, voice prompts, and vibrations. The multi-modal output allows users to intuitively recognize their current situation and, if necessary, view more information or modify their route on their mobile device screen. For example, as a user moves toward a subway station entrance, the system can provide real-time directional guidance, and prior to arrival, voice prompts such as "entrance 10 meters ahead on the left" to minimize disorientation.

The mobile terminal also works in conjunction with the XR glasses as an additional input and output device for the user. Users can change settings on the mobile device or control personalized services through the interface, and if necessary, send location and status information to a remote assistant through the emergency call function. This can also be utilized as a face-to-face response in emergency situations.

The server provides metadata based on public and on-site data and provides various information requested by the XR glasses and mobile devices in real time, such as the location of transportation facilities, arrival times, and route guidance. The server also continuously updates and manages data related to the user's travel route to help the user reach their destination safely and quickly.

With such a system structure, the XR system proposed in this research can integrate various functions such as indoor and outdoor transportation guidance, digital communication support, and contactless assistant services. Users can wear XR glasses and utilize mobile terminals as needed to receive information in a more effective and intuitive way, which is expected to overcome the limitations of mobility and communication and greatly improve their ability to live independently.

IV. ORGANIZATION AND ROLE OF AI IN THE SYSTEM

Artificial intelligence plays a key role in the XR system proposed in this research. AI is responsible for analyzing and processing the video data acquired by the XR glasses to provide users with the information they need in real time.

The overall structure and data flow of the XR system can be clearly seen in Figure 1. The XR glasses collect real-time video data by photographing the surrounding environment and transmit this data to a mobile terminal connected by a wire. An artificial intelligence model processes the video on the mobile device to perform object recognition and environment recognition, identifying key objects such as obstacles, pedestrians, vehicles, and traffic signs in real time. The server provides auxiliary information related to the user's movement route, accumulates detection results over the long term, and links them to customized services.

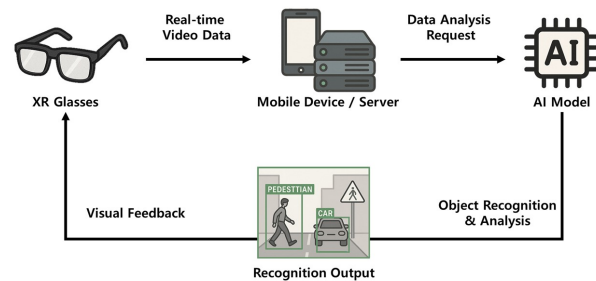


Fig. 1. Overall structure and data flow of the XR-based AI object detection system.

AI object detection technology mainly uses deep learning-based YOLO (You Only Look Once) models, which have a good balance between object detection accuracy and computation speed, making them suitable for wearable environments. In particular, the YOLOv8 model is lightweight and can be processed in real time even in mobile environments, and has the advantage of excellent detection performance for various object classes. These characteristics work very well in an XR-based wearable assistive device environment where real-time and resource constraints are critical, and are highly technically compatible with the mobile-embedded interworking structure of this research.

The system goes beyond simply recognizing objects and comprehensively analyzes the location and size of the detected object, its distance from the user, and the direction of movement to determine risk factors. For example, if a vehicle and a scooter are simultaneously detected in the direction in front of the user, and the object is approaching the user, the system determines that it is a “caution” level hazard and immediately displays a message to adjust the user's walking direction.

After object recognition, the system quantitatively assesses the level of risk based on attribute information such as the on-screen position coordinates and relative size of the detected object. The risk assessment is performed by combining a distance-based risk function with object type-specific weights, and the result is categorized as ‘no risk’, ‘caution’, ‘high risk’, etc. For example, a scooter approaching within two meters of the user will have a higher risk score than a stationary structure, which is used to prioritize notifications and adjust the intensity of the warning. This quantified risk determination ensures that users are notified in a more timely and relevant manner.

Table 1 below shows the main types of objects that the object recognition AI model used in this system can detect.

TABLE I. OBJECT CLASSIFICATION

Object Category	Subcategory	Objects	
Obstacle	Dynamic Obstacles	Car	Bus
		Truck	Bicycle
		Scooter	Motorcycle
	Static Obstacles	Bollard	Construction Sign
Pedestrian		Person	
Traffic Facility	Navigation Infrastructure	Crosswalk	Traffic Light
	Public Transportation	Bus Stop	Subway Station

Based on these object detection results, the mobile terminal and the server provide users with information such as real-time notifications, danger warnings, and route guidance. In addition, the server continuously stores and manages object detection results and movement data, which is used to analyze the user's movement habits and provide customized services according to the situation.

In this way, the AI technology in this research's XR system goes beyond simple detection to include situational understanding and response, and plays a crucial role in providing accurate and real-time information to users, enabling the elderly and disabled to move more safely and independently.

V. SYSTEM UTILIZATION SCENARIOS AND CASES

This chapter describes how the proposed XR-based AI system is utilized in real-world user environments through specific scenarios and examples. The system is designed to work effectively in three main situations: outdoor walking guidance, indoor mobility assistance, and digital communication assistance.

A. Outdoor walking assistance scenario

When a blind or visually impaired person is walking in a city, the XR glasses analyze the surrounding environment in real time based on the images acquired by the camera and recognize obstacles, crosswalks, vehicles, etc. on the road. The AI synthesizes the current location and orientation, as well as the movement and distance of the detected objects, and provides voice or visual alerts to help the user navigate safely.

Through in-system preferences, users can adjust the guidance method (voice/vibration/visualization), feedback intensity, notification frequency, and more. These customizable feedback settings provide an immersive walking experience and allow users to receive environmental information in a way that is appropriate for their preferences and disabilities.

For example, if you are stopped in front of a crosswalk and a vehicle is detected within 5 meters in front of you that is about to pass without stopping, the system will warn you with a “Vehicle approaching, be careful” alert. Conversely, if the vehicle is stopped and the light has turned green, the system switches to “Cross now”. In addition, when two or more types of obstacles are detected ahead (e.g., scooter + construction sign), the system adjusts the intensity or frequency of the warning voice according to the relative risk, helping users intuitively understand which direction to avoid first.



Fig. 2. XR-based outdoor walking guidance interface with obstacle and vehicle detection.

B. Supporting access to indoor transportation facilities

The system is also useful in complex indoor transportation spaces such as subway stations and terminals. By detecting major means of transportation such as elevators, escalators, signs, and destinations, the system provides step-by-step guidance to help users reach their destination independently. Since GPS signals are limited, especially indoors, the system relies on visual sign recognition and tracking of user movement trajectories to improve route guidance accuracy.

For example, if a visually impaired person wants to use an elevator to get to a subway platform, the system can say “Elevator is 12 meters on the left. Go straight and turn right.” The system provides specific instructions such as “Elevator is 12 meters on the left. If the elevator is inaccessible due to crowds or obstacles, the system can also provide rerouting flexibility by automatically suggesting alternative escalator directions. When integrated with real-time crowding

forecasting, users can also be provided with additional information such as “Exit 1 is currently crowded”.



Fig. 3. Indoor navigation support for visually impaired users in subway stations.

VI. CONCLUSION AND FUTURE RESEARCH CHALLENGES

In this paper, we discussed the concept, configuration, and practical application of an XR-based AI support system for the elderly and disabled. By integrating indoor and outdoor traffic guidance, hazard recognition, and remote assistant connectivity, this system has the potential to secure users' mobility and safety at the same time.

The current study focused on system configuration and implementation of basic AI functions, but future research should include more quantitative and customized indicators. In particular, the development of an AI-based safe walking index calculation algorithm is one of the key future tasks. This index quantifies the risk level of the current walking environment by synthesizing object recognition results, environmental information, user location, and movement patterns, and can provide users with more intuitive and reliable route guidance.

Another important research direction is the enhancement of the obstacle alarm function using AI. Beyond simple detection, it is required to quantitatively analyze the risk by comprehensively determining the direction of movement of the object, distance from the user, and relative risk, and deliver the results to the user through a voice or visual interface. The risk level is not a one-time judgment, but is updated in real-time to reflect the continuous movement of objects, changes in position, speed, etc. This allows users to receive dynamic feedback that adapts to changing conditions rather than static warnings, enabling more reliable walking in complex urban environments.

There is also the potential for the system to learn from long-term accumulated user data and extend to personalized guidance that reflects recurring travel patterns or route preferences. For example, if a user's frequent transportation entrances and exits, types of obstacles they avoid, and preferred feedback methods are accumulated, AI could analyze this data to optimize more personalized route guidance and warnings in the future. This direction can also be developed into a real-time adaptive AI structure based on reinforcement learning, which will be a key technology to continuously improve the user experience.

In addition, non-face-to-face interaction with remote assistants is also an important future enhancement. Currently, it is centered on basic vision sharing and voice interfaces, but future research needs to improve the stability of real-time video transmission, automatic detection and categorization of emergency situations, and two-way interaction functions that complement user expression. In particular, an extension design that considers the connection with care services, guardian monitoring systems, etc. is also required.

If these studies are continued, the proposed XR system is expected to contribute not only to technical perfection but also to social effectiveness and utilization. In particular, considering the increasing demand for assistive technology in an aging society and the need for policies to realize a digitally inclusive society, the technology in this study can be established as an important foundation technology.

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